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Theory and Numerical Method: LES-PISO Flow Past a Circular Cylinder at $Re = 3900$

This document describes the mathematical model and numerical algorithm used in the `cylinder_re3900` simulation. It is written so that an independent implementation can be reproduced from the theory alone. No source code is included.

1. Problem statement

1.1 Physical problem

Simulate three-dimensional, unsteady, incompressible turbulent flow of a Newtonian fluid past a circular cylinder of diameter D at Reynolds number

$$Re_D = \frac{U_\infty D}{\nu} = 3900,$$

where U_∞ is the freestream speed and ν is the kinematic viscosity. This Reynolds number lies in the shear-layer transition regime: the wake is turbulent and three-dimensional, while the attached boundary layers on the cylinder remain laminar. Large-eddy simulation (LES) is used to resolve the large energetic eddies and to model the effect of unresolved scales through an eddy viscosity.

1.2 Computational domain

The fluid occupies a rectangular box

$$\Omega = [0, L] \times [0, W] \times [0, H].$$

Default geometric choices (in units of D):

Quantity	Symbol	Typical value	Role
Streamwise length	L	$20 D$	inlet -> outlet
Cross-stream width	W	$12 D$	far-field y
Spanwise height	H	πD	cylinder axis / span
Cylinder centre	(c_x, c_y)	$(5D, 6D)$	axis parallel to z
Cylinder radius	$R = D/2$	$0.5 D$	solid obstacle

The cylinder is an infinite (extruded) circular cylinder: every (x, y) inside the disk of radius R is solid for all z .

1.3 Inputs that fully define a run

A complete case is specified by:

1. **Physical parameters:** density ρ , freestream speed U_∞ , diameter D , Reynolds number Re_D (hence $\nu = U_\infty D / \text{Re}_D$).
2. **Domain:** L, W, H and cylinder centre (c_x, c_y) .
3. **Grid:** integers n_x, n_y, n_z (number of pressure cells in each direction).
4. **Time:** step Δt , end time t_{end} (often reported as convective time $t U_\infty / D$).
5. **LES:** Smagorinsky constant C_s (default 0.1), on/off flag.
6. **PISO:** number of pressure-velocity correctors N_c (default 2).
7. **Initial fields:** usually uniform freestream $u = \bar{U}_\infty, v = w = 0, p = 0$, plus a tiny random perturbation on v to break symmetry so vortex shedding can start.
8. **Outputs:** force coefficients, mid-span velocity/vorticity visualisations, optional animation.

With the nondimensional choice $D = 1, U_\infty = 1$, one has $\nu = 1/3900$.

2. Governing equations: incompressible Navier-Stokes

Before any LES modelling, the continuum equations for a constant-density Newtonian fluid are the incompressible Navier-Stokes (NS) equations.

2.1 Continuity (mass)

$$\nabla \cdot \mathbf{u} = 0, \quad \mathbf{u} = (u, v, w).$$

2.2 Momentum

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u}.$$

Component form:

$$\begin{aligned} \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} &= -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \nabla^2 u, \\ \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} &= -\frac{1}{\rho} \frac{\partial p}{\partial y} + \nu \nabla^2 v, \\ \frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} &= -\frac{1}{\rho} \frac{\partial p}{\partial z} + \nu \nabla^2 w. \end{aligned}$$

Why these equations? Constant density and low Mach number justify incompressibility. Pressure is not thermodynamic; it is a Lagrange multiplier that enforces $\nabla \cdot \mathbf{u} = 0$.

What is unknown? The velocity $\mathbf{u}(\mathbf{x}, t)$ and pressure $p(\mathbf{x}, t)$.

3. Large-eddy simulation (LES) and the Smagorinsky model

3.1 Why LES?

Direct numerical simulation (DNS) of all scales at $\text{Re} = 3900$ is expensive. LES resolves only scales larger than a filter width Δ comparable to the grid, and models the effect of smaller eddies on the resolved motion.

Formally, a spatial filter $(\overline{\cdot})$ applied to the NS equations yields equations for the resolved velocity $\overline{\mathbf{u}}$. The nonlinear term produces an unclosed residual stress

$$\tau_{ij} = \overline{u_i u_j} - \overline{u_i} \overline{u_j},$$

which must be modelled.

3.2 Eddy-viscosity closure

The Smagorinsky model treats the residual stress like a Newtonian viscous stress with an eddy (turbulent) viscosity ν_t :

$$\tau_{ij} - \frac{1}{3} \tau_{kk} \delta_{ij} \approx -2\nu_t \overline{S}_{ij},$$

where the resolved strain-rate tensor is

$$\bar{S}_{ij} = \frac{1}{2} \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right).$$

Its magnitude is

$$|\bar{S}| = \sqrt{2 \bar{S}_{ij} \bar{S}_{ij}}.$$

3.3 Smagorinsky eddy viscosity

$$\nu_t = (C_s \Delta)^2 |\bar{S}|.$$

The filter width on a Cartesian cell is taken as the geometric mean of the spacings,

$$\Delta = (\Delta x \Delta y \Delta z)^{1/3}.$$

The constant C_s is an input (here $C_s = 0.1$ for the cylinder case).

3.4 Effective viscosity in the momentum equation

Substituting the model into the filtered momentum equation gives the same form as NS, but with molecular viscosity replaced by

$$\nu_{\text{eff}} = \nu + \nu_t.$$

Thus the resolved equations solved numerically are

$$\begin{aligned} \nabla \cdot \bar{\mathbf{u}} &= 0, \\ \frac{\partial \bar{\mathbf{u}}}{\partial t} + (\bar{\mathbf{u}} \cdot \nabla) \bar{\mathbf{u}} &= -\frac{1}{\rho} \nabla \bar{p} + \nabla \cdot (\nu_{\text{eff}} \nabla \bar{\mathbf{u}}) \approx -\frac{1}{\rho} \nabla \bar{p} + \nu_{\text{eff}} \nabla^2 \bar{\mathbf{u}}, \end{aligned}$$

where the last step uses the implementation's constant-within-a-cell treatment of ν_{eff} in the viscous term (Laplacian with face-interpolated ν_{eff}).

Overlines are dropped below: $\mathbf{u}$ and p mean **resolved** LES fields.

3.5 Discrete evaluation of $|\bar{S}|$ on the staggered grid

Velocities live on faces (Section 4). Strain components are **not** formed by first averaging all velocities to cell centres.

Normal strains at pressure (cell) centres from face jumps:

$$S_{xx} = \frac{u_{i+1/2} - u_{i-1/2}}{\Delta x}, \quad S_{yy} = \frac{v_{j+1/2} - v_{j-1/2}}{\Delta y}, \quad S_{zz} = \frac{w_{k+1/2} - w_{k-1/2}}{\Delta z}.$$

Shear strains: evaluate $\partial u/\partial y$, $\partial u/\partial z$ on u -faces (central differences in the face-tangential directions; one-sided at domain edges), and likewise $\partial v/\partial x$, $\partial v/\partial z$ on v -faces and $\partial w/\partial x$, $\partial w/\partial y$ on w -faces. Average those derivative fields from faces to the adjacent cell centre, then

$$S_{xy} = \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)_c, \quad S_{xz} = \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right)_c, \quad S_{yz} = \frac{1}{2} \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right)_c.$$

Magnitude:

$$|S| = \sqrt{2(S_{xx}^2 + S_{yy}^2 + S_{zz}^2) + 4(S_{xy}^2 + S_{xz}^2 + S_{yz}^2)}.$$

Then $\nu_t = (C_s \Delta)^2 |S|$ and $\nu_{\text{eff}} = \nu + \nu_t$ are stored at cell centres. For the viscous term on faces, ν_{eff} is linearly interpolated from the two adjacent cells onto each velocity face.

Stability clipping (implementation detail that must be reproduced): ν_{eff} is limited from below by ν , from above by a multiple of ν , and also by an explicit viscous stability bound of the form

$$\nu_{\text{eff}} \leq \frac{C_\nu}{\Delta t (\Delta x^{-2} + \Delta y^{-2} + \Delta z^{-2})},$$

with $C_\nu \approx 0.15$ in this solver, so that the explicit viscous update remains stable.

4. Spatial discretisation: MAC staggered Cartesian grid

4.1 Why staggered?

On a collocated grid, central pressure gradients and velocity divergences can admit odd-even decoupling (checkerboard pressure). The Marker-and-Cell (MAC) staggered arrangement places each velocity component on the face normal to that component and pressure at the cell centre. Then:

- $\nabla \cdot \mathbf{u}$ at centres uses face fluxes directly;
- ∇p on faces uses neighbouring cell pressures;

so the discrete gradient and divergence are adjoint (up to boundary terms), which strongly reduces checkerboarding.

4.2 Mesh

Divide $[0, L] \times [0, W] \times [0, H]$ into $n_x \times n_y \times n_z$ equal cells:

$$\Delta x = \frac{L}{n_x}, \quad \Delta y = \frac{W}{n_y}, \quad \Delta z = \frac{H}{n_z}.$$

Cell-centre coordinates:

$$x_i = (i + \frac{1}{2})\Delta x, \quad y_j = (j + \frac{1}{2})\Delta y, \quad z_k = (k + \frac{1}{2})\Delta z,$$

with $i = 0, \dots, n_x - 1$, etc.

Face coordinates:

- u -faces at $x = i\Delta x$, $i = 0, \dots, n_x \rightarrow$ array size $(n_x + 1) \times n_y \times n_z$;
- v -faces at $y = j\Delta y$, $j = 0, \dots, n_y \rightarrow$ size $n_x \times (n_y + 1) \times n_z$;
- w -faces at $z = k\Delta z$, $k = 0, \dots, n_z \rightarrow$ size $n_x \times n_y \times (n_z + 1)$;
- pressure / ν_{eff} : size $n_x \times n_y \times n_z$.

4.3 Discrete divergence (cell centres)

$$(\nabla \cdot \mathbf{u})_{i,j,k} = \frac{u_{i+1,j,k} - u_{i,j,k}}{\Delta x} + \frac{v_{i,j+1,k} - v_{i,j,k}}{\Delta y} + \frac{w_{i,j,k+1} - w_{i,j,k}}{\Delta z}.$$

Inside solid cells this is set to zero (immersed boundary).

4.4 Discrete pressure gradient (faces)

For interior u -faces $i = 1, \dots, n_x - 1$:

$$\left(\frac{\partial p}{\partial x}\right)_{i,j,k} = \frac{p_{i,j,k} - p_{i-1,j,k}}{\Delta x},$$

and analogously for v - and w -faces. Boundary faces that carry Dirichlet velocity are not corrected by pressure in the same way (inlet u is prescribed).

4.5 Convection: first-order upwind

For a scalar ϕ advected by a velocity component c along a grid direction with spacing h ,

$$c \frac{\partial \phi}{\partial \xi} \approx \begin{cases} c(\phi_m - \phi_{m-1})/h & c \geq 0, \\ c(\phi_{m+1} - \phi_m)/h & c < 0. \end{cases}$$

Why upwind? It is dissipative and stabilises explicit advection at finite CFL. **Trade-off:** more numerical diffusion than central or higher-order schemes.

On staggered faces, the three advecting velocities at a u -face are: u itself, and v, w interpolated from neighbouring v - and w -faces to that u -face (and cyclic permutations for v - and w -momentum).

The convective residual for u is

$$((\mathbf{u} \cdot \nabla)u) = u\partial_x u + v_u\partial_y u + w_u\partial_z u,$$

with v_u, w_u the interpolants onto the u -face.

4.6 Viscous term: seven-point Laplacian

On each velocity component's staggered array,

$$\nabla^2 \phi \approx \frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{\Delta x^2} + \frac{\phi_{j+1} - 2\phi_j + \phi_{j-1}}{\Delta y^2} + \frac{\phi_{k+1} - 2\phi_k + \phi_{k-1}}{\Delta z^2},$$

with one-sided (copy / Neumann-like) formulae at the array edges. Multiplied by the face value of ν_{eff} .

4.7 Explicit momentum residual without pressure

Define, on each face family,

$$\mathbf{H}(\mathbf{u}) = -(\mathbf{u} \cdot \nabla)\mathbf{u} + \nu_{\text{eff}}\nabla^2\mathbf{u}.$$

Pressure is treated separately in the PISO projection (Section 6).

5. Time integration: explicit or implicit?

5.1 Answer in one sentence

Momentum advection and diffusion are advanced explicitly (forward Euler using residuals at the known time level). **Pressure is obtained implicitly** by solving a Poisson equation each corrector so that the updated velocity is (approximately) divergence-free.

5.2 Explicit momentum step

Given fields at time t^n , the provisional velocity (predictor) is

$$\mathbf{u}^* = \mathbf{u}^n + \Delta t \left(\mathbf{H}(\mathbf{u}^n) - \frac{1}{\rho} \nabla p^n \right).$$

No linear system is solved for $\mathbf{u}^*$. That is classical **explicit** (forward Euler) treatment of $\mathbf{H}$.

Consequences:

- Time step is limited by CFL for advection,

$$\text{CFL} = \frac{U_{\max} \Delta t}{\min(\Delta x, \Delta y, \Delta z)} \lesssim O(1)$$

(a soft monitor value ≈ 0.5 is used as guidance), and by viscous stability when ν_{eff} is large (hence the ν_{eff} cap above).

- Implementation is simple; accuracy in time is first order.

5.3 Implicit pressure Poisson

Enforcing $\nabla \cdot \mathbf{u}^{n+1} = 0$ leads to

$$\nabla^2 p' = \frac{\rho}{\Delta t} \nabla \cdot \mathbf{u}^*,$$

solved as a sparse linear system (conjugate gradient, with a direct sparse fallback). This solve is **implicit** in p' . Velocity is then corrected explicitly using $\nabla p'$.

So the overall scheme is: **explicit momentum + implicit pressure projection** (a fractional-step / projection family method), organised as PISO.

6. PISO algorithm (Issa)

PISO (Pressure Implicit with Splitting of Operators) improves a single projection by repeating pressure-velocity corrections and, after the first corrector, rebuilding the explicit residual $\mathbf{H}$ from the latest velocity (neighbour / flux update).

Let N_c be the number of correctors (default 2).

6.1 Step A — LES viscosities

From $\mathbf{u}^n$, compute ν_{eff} at cell centres (Section 3) and interpolate to faces.

6.2 Step B — Momentum predictor

$$\mathbf{u}^* = \mathbf{u}^n + \Delta t \left(\mathbf{H}(\mathbf{u}^n) - \frac{1}{\rho} \nabla p^n \right),$$

then apply boundary conditions and the immersed-boundary condition (Section 7).

6.3 Step C — Corrector loop ($m = 1, \dots, N_c$)

1. **If** $m > 1$ (PISO neighbour update): recompute

$$\mathbf{u}^* = \mathbf{u}^n + \Delta t \left(\mathbf{H}(\mathbf{u}_{\text{prev}}^*) - \frac{1}{\rho} \nabla p \right)$$

using the latest corrected velocity inside $\mathbf{H}$, and re-apply BCs/IB.
(Here p is the pressure accumulated so far in the step.)

2. **Divergence:** $d = \nabla \cdot \mathbf{u}^*$ at cell centres (zero in solids).

3. **Pressure-correction Poisson:**

$$\nabla^2 p' = \frac{\rho}{\Delta t} d.$$

Discrete Laplacian: standard 7-point stencil with homogeneous Neumann treatment at domain faces (missing neighbour omitted). The null space of Neumann Poisson is removed by pinning one cell ($p' = 0$ at a reference cell) and removing the mean of the right-hand side. After the solve, subtract the mean of p' .

4. **Velocity correction** (on faces):

$$\mathbf{u}^* \leftarrow \mathbf{u}^* - \frac{\Delta t}{\rho} \nabla p'.$$

5. **Pressure update:**

$$p \leftarrow p + p'.$$

6. Re-apply BCs and IB to $\mathbf{u}^*$.

6.4 Step D — Accept new time level

$$\mathbf{u}^{n+1} = \mathbf{u}^*, \quad p^{n+1} = p - \langle p \rangle$$

(mean pressure set to zero; only gradients matter for incompressible flow).

6.5 Why PISO rather than a single projection?

A single fractional step uses one Poisson solve with $\mathbf{H}(\mathbf{u}^n)$ frozen. Extra correctors with an updated $\mathbf{H}(\mathbf{u}^*)$ reduce splitting error between the momentum residual and the divergence-free constraint, which is the defining idea of PISO relative to a pure projection method.

7. Boundary conditions and immersed cylinder

7.1 Far-field / domain boundaries

Location	Condition	Meaning
Inlet $x = 0$	$u = U_\infty, v = 0, w = 0$	uniform freestream
Outlet $x = L$	$\partial_x(u, v, w) = 0$ (copy from interior)	convective / Neumann outflow
$y = 0$ and $y = W$	$u = U_\infty, v = 0, \partial_y w \approx 0$	freestream / slip sides
$z = 0$ and $z = H$	$\partial_z u \approx 0, \partial_z v \approx 0, w = 0$	slip span ends (surrogate for periodic span)

These are **not** viscous channel walls; the setup is an external flow past a cylinder in a truncated box.

7.2 Immersed boundary (cylinder)

Solid cells: all pressure cells whose centre (x_i, y_j) satisfies

$$(x_i - c_x)^2 + (y_j - c_y)^2 \leq R^2$$

for every k .

Any velocity face that borders at least one solid cell is treated as solid and set to zero (stationary cylinder):

$$u = v = w = 0 \quad \text{on solid-adjacent faces.}$$

Divergence in solid cells is forced to zero so those cells do not drive the Poisson right-hand side.

This is a simple discrete-forcing / masking immersed-boundary approach: geometry is stair-stepped on the Cartesian mesh; fidelity improves as $\Delta x, \Delta y$ decrease relative to D .

7.3 Field sanitation

After updates, non-finite values are cleared and velocities are clipped to a large bound (order $8U_\infty$) to prevent rare blow-ups on coarse grids.

8. Force coefficients

Drag and lift coefficients are estimated from pressure on a thin band of **fluid** cells just outside the cylinder:

$$C_D = \frac{F_x}{\frac{1}{2}\rho U_\infty^2 D H}, \quad C_L = \frac{F_y}{\frac{1}{2}\rho U_\infty^2 D H}.$$

With outward unit normal in the xy -plane $\mathbf{n} = (n_x, n_y)$ from the cylinder centre through each band cell, and cell area scale $A \sim \min(\Delta x, \Delta y) \Delta z$,

$$\mathbf{F} \approx \sum_{\text{band}} (-p) A \mathbf{n}.$$

Viscous surface stress is **not** included in this estimate; it is a pressure-dominated approximation suitable for monitoring, not a high-precision force balance.

9. Initialisation

1. Set all u -faces to U_∞ , all v - and w -faces to 0, pressure to 0.
 2. Apply BCs and IB.
 3. Add a very small random perturbation to v (amplitude $\sim 10^{-3}U_\infty$) so that the symmetric wake can bifurcate into vortex shedding.
 4. Advance with the algorithm of Section 6 until $t = t_{\text{end}}$.
-

10. End-to-end algorithm (reproduction checklist)

For each time step Δt :

1. **Inputs known:** $\mathbf{u}^n, p^n$, grid, ρ, ν, C_s, N_c , solid mask.
2. **LES:** compute staggered $|S|, \nu_t, \nu_{\text{eff}}$ at centres; interpolate ν_{eff} to faces; apply stability clips.
3. **Predictor (explicit):** $\mathbf{u}^* = \mathbf{u}^n + \Delta t(\mathbf{H}(\mathbf{u}^n) - \nabla p^n / \rho)$; apply BC/IB.
4. **For** $m = 1$ to N_c :
 - if $m > 1$: rebuild $\mathbf{H}(\mathbf{u}^*)$, re-predict from $\mathbf{u}^n$ with current p , apply BC/IB;
 - form $d = \nabla \cdot \mathbf{u}^*$;
 - solve $\nabla^2 p' = (\rho / \Delta t) d$ (implicit);
 - $\mathbf{u}^* \leftarrow \mathbf{u}^* - (\Delta t / \rho) \nabla p', p \leftarrow p + p'$;
 - apply BC/IB.
5. **Store** $\mathbf{u}^{n+1}, p^{n+1}$; optionally compute C_D, C_L and mid-span diagnostics.

Time integration summary

Term	Treatment
Convection $(\mathbf{u} \cdot \nabla)\mathbf{u}$	Explicit (in $\mathbf{H}$)
Viscous $\nu_{\text{eff}} \nabla^2 \mathbf{u}$	Explicit (in $\mathbf{H}$)
Old pressure ∇p^n in predictor	Explicit
Pressure correction p'	Implicit Poisson
Overall	Explicit momentum + PISO projection

11. Quantities to monitor for a correct implementation

- **Mass conservation:** $\max |\nabla \cdot \mathbf{u}|$ in fluid cells should drop after each projection (boundaries/IB can leave local residuals).
- **CFL:** keep CFL moderate; reduce Δt if the solution diverges.
- **Wake physics at $\text{Re} = 3900$:** after a few convective times, expect a separated wake, shear layers, and (on adequate grids and longer times) unsteady shedding; mean C_D on fine LES grids is classically near order 1 (coarse grids over-predict drag).
- **Energy of ν_t :** ν_t should be small in quiet freestream and larger in shear layers/wake.

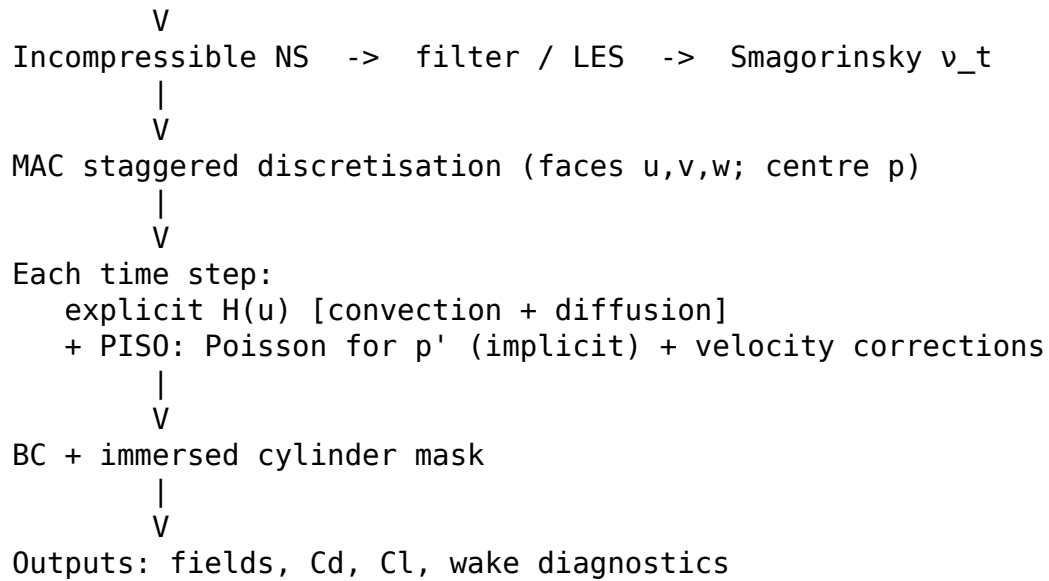
12. Default numerical parameters (reference case)

Parameter	Value
Re_D	3900
D, U_∞, ρ	1, 1, 1
ν	1/3900
Domain	$20 \times 12 \times \pi$ (in units of D)
Grid (production)	$128 \times 80 \times 24$
Δt	0.002
t_{end}	20 (convective times)
C_s	0.1
PISO correctors N_c	2
Convection scheme	1st-order upwind
Momentum time scheme	Forward Euler (explicit)
Pressure	Implicit Poisson (CG)

13. Conceptual flowchart

Inputs (Re , domain, grid, Δt , C_s , N_c)

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14. References (methods)

- Harlow & Welch (1965): staggered MAC grid for incompressible flow.
- Smagorinsky (1963): eddy-viscosity closure for unresolved scales.
- Issa (1986): PISO — pressure-implicit with splitting of operators.
- Standard projection / fractional-step ideas (Chorin; later variants): divergence-free enforcement via Poisson pressure.

This document matches the mathematical structure of the `cylinder_re3900` solver: filtered incompressible NS, Smagorinsky LES on a staggered strain-rate evaluation, explicit face momentum residuals, and an Issa-type PISO pressure-velocity coupling with an immersed circular cylinder at $Re = 3900$.